H Vasant Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.18
Sri Edu Foundation	Class XII (Karnataka State Board)	2024	93.83%
Vidya Bharati Public School	Class X (Central Board Of Secondary Education)	2022	98.40%

SCHOLASTIC ACHIEVEMENTS

- **IMC Prosperity 4 (Global Quant Trading Competition):** Secured AIR 3 (Global 36) in the penultimate round.
- **ICPC (International Collegiate Programming Contest):** Achieved a rank of **173** nationwide in preliminary round
- **IICPC Quantfest:** Achieved a rank 906 out of 10000+ participants.

SKILLS

- **Languages:**
- *Proficient* : C++, C
- *Working Knowledge* : Python (NumPy, Pandas, Matplotlib)
- *Familiar* : Java, Shell Scripting, Ocaml, Solidity
- **Query Processing Languages:** MySQL, PostgreSQL
- **Tools & Technologies:** Git, GitHub, Docker, CMake
- **Basic Web Development:** HTML, CSS, Basic Javascript

- **Core CS Concepts:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Operating Systems, DBMS

PROJECTS

RISC-V Pipelined Processor Simulator (C++) Github

- Extended an existing RISC-V simulator to support a **fully pipelined** processor with **multiple execution modes**.
- Implemented data hazard detection, stalling, and forwarding units for both **integer** and **floating-point** instructions.
- Integrated static and **dynamic branch prediction** using a Branch History Table (BHT) with configurable processor modes
- Analyzed performance by reporting total cycles, stalls, and flushes across different pipeline configurations.

Enhanced xv6 Operating System Subsystems (C) Github

- Developed a centralized page frame table with a stateful **Clock page replacement** algorithm
- Designed a System-Call-Aware **MLFQ scheduler** for dynamic process prioritization.
- Extended kernel process accounting to track system calls, memory usage, page faults, and swaps.
- Maintained multi-core correctness using kernel spinlocks and robust locking hierarchies.

Hybrid Core Scheduling for Performance on Linux (C, Group Project (2 members)) Github

- Co-developed an adaptive **thread-level scheduler** for hybrid Intel architectures to optimize workload distribution between P-cores and E-cores
- Built a real-time monitoring framework using Linux */proc* and turbostat to track CPU, **IPC**, **frequency**, **power**, and **temperature**.
- Implemented **affinity-based** thread migration with thermal safeguards and admission control for stable scheduling.
- Analyzed architectural trade-offs: Achieved a *~15%* average reduction in execution time at a *~7.1%* power cost, while

documenting a *~1%* throughput limit due to migration overhead

Arbitrary Precision Arithmetic Library (Java) Github

- Built an arbitrary-precision arithmetic library for integers and **floating-point numbers**.
- Implemented custom arithmetic algorithms without relying on built-in BigInteger.
- Designed reusable AInteger and AFloat classes using **object-oriented principles**.
- Automated testing with a Python harness and packaged the library as a reusable **JAR** using **Apache Ant**..

RELEVANT COURSES

- **Computer Science**: Data Structures, Algorithms, Computer Architecture, OS, DBMS ,Software Development Fundamentals, Theory of Computation
- **Mathematics:** Probability, Linear Algebra, Transform Techniques, Calculus

POSITION OF RESPONSIBILITY

Core Member, Infero (Algorithmic & Logical Club, IIT Hyderabad)

- Conducted DSA and competitive programming sessions while coordinating ACPL, the institute's flagship intra-college coding contest

Teaching Assistant, Introduction to Programming (IIT Hyderabad)

- Mentored freshmen in programming labs, assisting with debugging, evaluating assignments, and reinforcing core programming concepts

Hospitality Coordinator, Elan & nVision (IIT Hyderabad)

- Managed logistics, hospitality, and accommodation for external participants and guest artists during the flagship techno-cultural festival.

EXTRACURRICULAR

- **Competitive Programming:** Active and regular participant in high-stakes algorithmic contests on **Codeforces**; achieved a peak rating of **1493 (Specialist)** and successfully solved **650+ complex problems** across the platform.
- **Algorithmic Problem Solving:** Regularly practice advanced data structures and algorithms by solving diverse problem sets on **CSES** and **LeetCode.**
- **E-WASTE MANAGEMENT IDEATHON:** Participated in ELAN & nVision E-WASTE MANAGEMENT IDEATHON and secured **second** position.